Nelson Samayoa | Software Developer

www.**linkedin**.com/in/nelsonsamayoa
www.**github**.com/nel-sam
www.nel-sam.github.io/portfolio/
nelson.rs.07@gmail.com



Senior Full-Stack Software Engineer with 16+ years of experience designing and building scalable applications, developer platforms, and cloud-native systems. Expertise spans .NET, React, Angular, TypeScript, Azure, AWS, mobile development, distributed systems, and DevOps. Delivered products serving consumers, enterprise customers, and educational organizations, including public APIs, AI-powered learning applications, large-scale LMS platforms, developer tooling, and automation frameworks.

Experience leading SDET and automation engineering teams, building CI/CD infrastructure, and driving quality engineering initiatives. Comfortable owning the entire product lifecycle from architecture and UX through implementation, cloud deployment, monitoring, and long-term maintenance.

Native speaker of English and Spanish, and fluent in Japanese and French.

CORE TECHNOLOGIES

Frontend: React (React.js), Angular (Angular.js), TypeScript

Backend: .NET, C#, REST APIs, SQL Server, Node.js (Electron)

Mobile: React Native, Flutter, Ionic

Cloud: Azure, AWS, Event-Driven Architecture

DevOps & Quality: CI/CD, Test Automation

EDUCATION

Bachelor of Science Computer Science Oregon State University
Bachelor of Arts in Spanish Portland State University

WORK EXPERIENCE

Software Developer | Entacit LLC

Remote *Apr 2022 - Nov 2026*

Founding full-stack developer on an education platform spanning iOS, Android, web, and desktop using React Native, Angular, Electron, and .NET.

- **Mobile development:** Core developer on a React Native (TypeScript) app from its inception: data management, user sync via QR codes, deep links, iOS development/testing, and build pipeline setup (dev/staging/prod iOS schemes). Performed multiple React Native major upgrades, built custom components to replace unmaintained dependencies, a complete UI/UX redesign of the app, forced-update flow, parent/teacher/manager views, and cross-platform media playback reliability. Create automated Mobile UI tests using Maestro for smoke testing.

- **Desktop development:** Built and maintained the Electron (React + Node.js) application, including tabs support, reliable image rendering (FastImage), dependency upgrades, and robust handling of unsupported file types and external links.

- **Web development:** Developed major features on the Angular web app: QR code generation, CSV generation and upload (with data-integrity handling for empty values), a Video Assessments module including Safari support and audio/video device testing, quality evaluation reports with permission-gated access, tags system, and horizontal stepper/form-component library work. Performed Angular version upgrades and set up build pipelines.

- **AI development:** Using Azure's OpenAI and Speech APIs, helped create an AI chat bot designed for English language learning; also worked with Google's Gemini API and Rive animations during prototyping. Built a pronunciation component with recording-loss prevention, error indicators, and resume-attempt handling for speech evaluation.

- **Backend development:** Using C# (.NET / .NET Core), Entity Framework, and MS SQL, built REST API features including: student points system, bulk operations (session delete/cancel, student username updates, bulk tag and custom-field uploads), custom fields with full CRUD + CSV import/export, course cloning, email notifications with timezone handling, password policy and forgot-password/username flows, and a full-stack student/parent account-deletion flow supporting data-protection compliance.

Software Developer | Imagine Learning

Remote *May 2021 - Mar 2022*

C#, React, TypeScript, AWS, PostgreSQL, GitHub, Spinnaker, CSS/SASS, Jira.
Full-stack engineer on a Learning Management System (LMS) platform serving students, educators, and administrators. Developed learner-facing experiences, reporting tools, backend services, and cloud-native event-processing systems that supported course delivery, progress tracking, enrollment management, and educational analytics.

- **Web Development:** Developed React-based applications supporting students and educators within the LMS platform. Created a highly customized progress-reporting application featuring complex tabular data visualization, collapsible sections, dynamic layouts, and advanced styling to improve access to student performance data.

- **Performance Engineering:** Participated in a large-scale performance initiative, identifying and implementing frontend and backend optimizations that improved application responsiveness, scalability, and overall user experience across the platform.

- **Cloud Architecture & Distributed Systems:** Designed and implemented AWS-based event-driven systems for processing student and application events. Developed scalable architecture patterns that enabled asynchronous processing and integration between platform components.

Software Developer | Paylocity

Remote *Mar 2020 - May 2021*

C#, .NET, React, Angular, Ionic, TypeScript, RabbitMQ, AWS, Azure, SQL Server, TeamCity, Octopus Deploy.
Full-stack engineer on a Learning Management System (LMS) platform serving online learning experiences and course management. Contributed to web, mobile, backend, and distributed-system development supporting course discovery, registration, content delivery, and learner engagement. Worked across React, Angular/Ionic, .NET, RabbitMQ, SQL Server, AWS, and Azure while helping define the architecture for next-generation learning solutions.

- **Mobile Development:** Developed Angular + Ionic mobile application features including course catalog browsing, search functionality, multi-criteria filtering, and user experience enhancements designed to improve course discovery and learner engagement.

- **Web Development:** Developed and maintained user-facing learning experiences within the LMS platform using React and Angular. Built new user interfaces, enhanced existing workflows, and contributed to the Learning module within the broader Paylocity suite.

- **Backend & Distributed Systems:** Implemented new work processors and background services using RabbitMQ to distribute workloads and execute tasks in parallel, improving scalability and processing efficiency within the learning platform.

- **Architecture & System Design:** Authored architecture documentation for a cross-platform learning solution utilizing React web clients, Angular/Ionic mobile applications, C#/.NET APIs, SQL Server, RabbitMQ messaging, and AWS storage services. Collaborated on system design decisions spanning the full application stack.

Software Developer | DriveTime

Tempe, AZ *May 2018 - Feb 2020*

C#, .NET, Angular, Ionic, TypeScript, GraphQL, Azure Functions, Service Bus, REST APIs, Azure Active Directory (OpenID Connect), SQL Server, Azure DevOps.
Full-stack engineer responsible for customer-facing web and mobile applications, public APIs, cloud-native Azure solutions, and authentication architecture. Contributed to DriveTime.com and Bridgecrest.com, helped establish frontend standards for Angular applications, and designed scalable event-driven systems using Azure services. Participated in architecture and security initiatives, including public API authentication and enterprise identity modernization through Azure Active Directory.

- **Web Development:** Full-stack development for DriveTime.com and Bridgecrest.com using Angular, TypeScript, C#, and .NET technologies. Introduced Angular Material into an existing Angular application and established UI/UX, performance, and code-quality standards. Contributed to multiple web applications and maintained legacy jQuery and KnockoutJS systems while supporting modernization efforts.

- **Mobile Development:** Developed hybrid mobile applications using Ionic and Angular. Independently recreated a legacy application in Angular as a modernization initiative, improving maintainability and aligning the platform with current frontend development practices.

- **API & Backend Development:** Contributed to the design and implementation of a public-facing GraphQL API built with C#/.NET for external consumers. Participated in architecture discussions, developed proof-of-concepts, and implemented the API authentication model. Developed REST-based services and backend functionality supporting customer-facing applications.

- **Cloud Architecture & Distributed Systems:** Architected and implemented event-driven Azure solutions using Azure Functions, Service Bus, Queues, Topics, Blob Storage, and Azure Active Directory. Designed systems for automated vehicle-image processing and vehicle-value analysis, leveraging serverless and message-driven architectures to improve scalability and reliability.

- **Identity & Security:** Led the migration of legacy products to Azure Active Directory authentication using OpenID Connect. Contributed to authentication and authorization architecture for externally consumed APIs, helping improve security and identity management across platforms.

Software Developer Engineer in Test (SDET) Team Lead | iPro Tech

Tempe, AZ *Jan 2017 - Mar 2018*

C#, Selenium, SpecFlow, PowerShell, Azure DevOps (VSTS/TFS), JavaScript.
Technical lead for automation engineering, responsible for team leadership, framework architecture, and test infrastructure.
Designed and maintained an enterprise automation platform used across development teams, including CI/CD integration, reporting systems, distributed test execution, and developer-facing tooling. Awarded three times for performance and earned two product certifications within two months of joining the company.

- **Test Automation Leadership:** Led an automation engineering team, managing sprint planning, delivery, and team priorities using TFS and Jira within Agile/Scrum processes. Directed the modernization of the company's automation strategy, including a large-scale migration to SpecFlow BDD and the White UI automation framework, significantly reducing tooling and maintenance costs while improving test maintainability.

- **Automation Framework Development:** Designed and implemented core features for a C#-based test automation platform, including Entity Framework-backed test-result logging, reporting infrastructure, asynchronous multi-browser execution, and distributed test execution across multiple agents. Improved scalability and execution efficiency for automated testing across product teams.

- **Internal Tools & Applications:** Developed internal support and reporting applications using C#, WinForms, JavaScript, and Team Services APIs. Built a multithreaded diagnostic utility used by

support teams to troubleshoot customer environments and created a web-based dashboard that visualized test health and quality metrics using Google Charts and TFS data.

- **DevOps & Infrastructure:** Managed build and test infrastructure using Team Foundation Server (TFS) and PowerShell automation. Maintained automated test environments and deployment processes to support continuous validation efforts across multiple development teams.

Software Developer & SDET Team Lead | Dell Inc

Remote *Jul 2012 - Jan 2017*

C#, ASP.NET MVC (Razor), WinForms, WCF, REST APIs, Entity Framework, SQL Server, Active Directory/LDAP, Azure DevOps (VSTS/TFS), PowerShell.
Progressed from Software Engineer to Automation Engineering Lead, developing enterprise applications, developer tooling, CI/CD infrastructure, and automated testing systems. Led a five-engineer team, mentored developers on testing and quality practices, and supported multiple product organizations through DevOps and automation initiatives. Worked on a fully distributed remote team and received the Dell Champion Award in 2015.

- **Desktop & Enterprise Application Development:** Developed and maintained internal and customer-facing applications using C#, WinForms, ASP.NET, IIS, Windows Services, and Web Services. Built a support-focused command-line utility that collected diagnostic information from Windows, SQL Server, and Active Directory environments to streamline customer issue investigation. Developed a WinForms reporting application for viewing historical security-policy snapshots through a REST API and later supported customers during the product's sunset phase through bug fixes, maintenance, and codebase training.

- **Test Automation Leadership:** Led the automated test development team consisting of five engineers, managing sprint planning, delivery, and stakeholder reporting using Azure DevOps (formerly VSTS). Established and expanded automated testing practices across product teams, including UI automation, API testing, load testing, and security-focused testing initiatives.

- **Developer Enablement & Training:** Mentored developers and test engineers on automation frameworks and quality engineering practices, delivering training on Microsoft CodedUI, Selenium, API automation, performance testing, and security and penetration testing methodologies.

- **DevOps & Continuous Integration:** Owned the build and test infrastructure supporting multiple product teams. Designed and maintained CI/CD systems, build servers, automated test environments, PowerShell-based automation, on-premises TFS/VSTS integrations, and Azure Virtual Machine environments used for automated validation and release processes.

Software Developer Intern | Intel Inc

Hillsboro, OR *Sep 2010 - Jul 2012*

C#, ASP.NET MVC (Razor), JavaScript, XML Processing, SSH Integration, Automation Frameworks.
Software engineer responsible for the development of an internal IDE and enterprise test automation platform. Built developer-facing tools, framework extensions, device-integration plugins, and web-based management applications while serving as a technical resource for SDETs and engineers. Regularly trained users, resolved automation challenges, and improved the scalability and reliability of testing systems.

- **Developer Tools & IDE Development:** Built and maintained an internal IDE used by SDETs to develop and debug automated tests. Implemented support for processing XML files larger than available RAM, persistent breakpoint storage, and JavaScript language tooling including syntax highlighting and autocomplete.

- **Test Automation Infrastructure:** Extended the automation platform through custom plugins and framework integrations, including Linux SSH command execution, Wake-on-LAN support, battery-state monitoring via WMI, and automated Cisco access point testing through an SSH abstraction layer.

- **Web Applications:** Developed an ASP.NET (Razor) web interface for remotely managing automation runs, execution environments, and framework operations.

- **Automated Testing:** Developed and maintained automated test suites for Bluetooth-enabled devices, leveraging the internal automation framework to create reliable, repeatable validation of device functionality and connectivity scenarios. Worked directly with SDETs to troubleshoot test failures, improve test reliability, and enhance framework capabilities.

PERSONAL PROJECTS

Language Learning Mobile Applications in Flutter

Designed, developed, and published three cross-platform grammar-learning applications for Japanese, Korean, and Serbo-Croatian on the Apple App Store and Google Play Store.

- Led product direction, feature design, and UI/UX strategy
- Partnered with a designer to create a polished and intuitive learning experience
- Mentored a junior developer who contributed to implementation
- Built the applications using Flutter with a lightweight offline-first architecture that avoided backend dependencies, enabling fast performance and simplified maintenance
- Managed the complete software lifecycle including requirements, development, testing, app store releases, user feedback, and ongoing updates.